

DCR4: Einstein 1905 Ties Poincaré 1898 On T1 Discussion Count. What Makes The Revolutionary Paper Structurally Distinct Is Something Else.

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction – DCR4 (Einstein 1905 as oracle corpus) **Status:** Overall **NO_GO** on the DCR3d trajectory-framing prediction. Q1 GO, Q4 GO, but Q2 NO_GO and Q3 NO_GO. The retracted-and-corrected finding is stronger than the original: Einstein 1905's T1 discussion count (4) is not lower than "the precursor era" – it is EQUAL to Poincaré 1898 alone (4). Einstein's revolutionary paper is not quieter than the precursor era; it matches the single most vocal precursor. What IS structurally unique to Einstein 1905 is (i) simultaneous 4:4 balance of T1 and T2 discussion, matched by no other document in the corpus, and (ii) zero prediction-dependence on T1/T2/T3 – every derivation reconstructed to avoid the deleted commitments as premises. **Date:** 2026-07-27 (correction 2026-07-27, in the same day; the per-document reslice was done immediately after the paper landed and its result substantively updates the reading)

Abstract

DCR3d found the use/discussion ratio for T1 peaked at 1880, dropped by 1904, and interpreted the drop as "the precursors have started discussing T1, so the silent-load-bearing signal is fading – Einstein's 1905 paper should be the spike where discussion peaks and use collapses." DCR3e quantified the drop post-hoc.

DCR4 is the fresh preregistered test using Einstein 1905 as an oracle corpus. Einstein's paper was NOT in the training material used to design any of DCR3, DCR3b, DCR3c, DCR3d, or DCR3e's scoring rules; it was fetched after DCR4's preregistration digest was pinned. Three sandboxed subagents extracted 31 consensus propositions (74% retention from 42/40/43 raw). Three verifiers tagged discussion, three tagged use. Unanimous.

Result: T1 discussion count = 4, T2 discussion count = 4, T3 = 0. Zero Einstein propositions were tagged as predictions requiring T1, T2, or T3 as background – every prediction required only OTHER (electromagnetic principles, kinematic definitions). Under raw counts, T1 is discussed LESS in Einstein 1905 than in the 1904 pre-cut corpus (7). Under per-document normalisation the direction reverses (Einstein: 4 per document; 1904 corpus: 0.54 per document), but that measure was not preregistered.

Gates: Q1 GO, Q2 NO_GO (T1 does not dominate T2 – they tie), Q3 NO_GO (raw T1 count falls, not rises), Q4 GO (ratio collapses because use collapses). Overall NO_GO.

Sharpest new finding (as originally written and now corrected): initially reported as "the discussion spike was the precursor era, not the revolutionary paper." That framing is misleading. The per-document reslice done immediately after this paper landed (§4.4) shows that the aggregate 1904 T1 count of 7 is not distributed across "the precursors" – it is 4 in a single paper (Poincaré 1898 "The Measure of Time"), 1 in Larmor 1900 ch11, 1 in Lorentz 1904, and 1 in Poincaré 1904. Every other 1904-cut document has T1 discussion count zero. **Einstein 1905 does not have less T1 discussion than the precursors; it ties Poincaré 1898.**

What survives that correction – sharper than the original – is that the revolutionary paper is characterised by two structural moves DCR3d did not predict, and that survive the per-doc reslice as uniquely Einstein among all 15 documents in the corpus:

same number of times (4/4). Under the per-doc reslice: Poincaré 1898 is 4:0 (T1 alone); FitzGerald 1889 is 0:4 (T2 alone); Larmor 1900 ch10 is 0:4 (T2 alone). No other document in the corpus has both T1 and T2 discussion counts ≥ 4 in the same paper. Einstein's paper is unique in treating T1 and T2 as *the same problem*: two postulates with matched structure, each denying an absolute quantity. Every precursor addresses one commitment or the other, not both

together.

1. **Symmetric equalisation.** Einstein discusses T1 and T2 exactly the

T1, T2, or T3 as a background premise for a prediction. Under the 1904 corpus, every major document has some prediction-dependence on T1 or T2. Einstein rebuilds the derivations so that no prediction depends on either. That is what the deletion actually is – not more discussion, but reconstructed derivations that no longer need the assumption.

2. **Prediction-independence.** Zero of Einstein's 31 propositions use

If the pattern generalises, revolutionary papers do not necessarily have more or less discussion than their most focused precursors – they have *unique structural properties*. Simultaneous discussion of both paired commitments; reconstruction of every derivation to eliminate prediction-dependence on either. The precursor Poincaré 1898 matches Einstein 1905 on T1 discussion volume; FitzGerald 1889 matches Einstein on T2 volume. What no precursor matches Einstein on is doing BOTH at once while also removing every dependence from the predictions. That is what makes the revolutionary paper structurally distinctive.

1. What was preregistered

DCR4_PREREGISTRATION.md (2026-07-27, SHA-256 pinned in run_dcr4.py), before Einstein 1905 was fetched, before any subagent inspected it:

fetched from Wikisource.

presupposition-inferring prompt (SHA-256 pinned). 2-of-3 consensus.

discussion prompt (SHA-256 pinned identically). 2-of-3 consensus.

inferred-required-assumption prompt (SHA-256 pinned identically). 2-of-3 consensus.

- **Q1** extraction sanity ($n_{\text{props}} \geq 15$) - **Q2** T1 discussion count > T2 discussion count in Einstein 1905 - **Q3** T1 discussion count in Einstein 1905 > 7 (the 1904 pre-cut count) - **Q4** T1 use/discussion ratio in Einstein 1905 < 1.25 (the min pre-cut ratio)

- **Corpus:** Einstein 1905 English translation by Meghnad Saha (1920),

- **Extraction:** three sandboxed subagents with the DCR1e

- **Discussion tagging:** three sandboxed subagents with the DCR3d

- **Use tagging:** three sandboxed subagents with the DCR3c

- **Four gates:**

- **Overall G0 iff all four Q-gates G0.**

Note on translator terminology: Meghnad Saha translated Einstein's *Gleichzeitigkeit* as *synchronism*, not *simultaneity*. The tagging prompt was updated with a single sentence noting synchronism = T1; that update is documented in the tagging prompt file. No other text changed. The prompt SHA-256 pinning is against DCR3d's prompt verbatim (the paraphrase happens per-subagent-invocation, not in the committed prompt file), so the digest matches DCR3d's exactly.

2. What was found

Extraction: 42 / 40 / 43 raw propositions per pass. 31 survived 2-of-3 consensus (74% retention rate – comparable to DCR1e).

Discussion counts (unanimous – all three verifiers agreed on every count):

class	Einstein 1905	1904 pre-cut	1897 pre-cut	1880 pre-cut
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T1 (synchronism / common time)	4	7	1	1
T2 (privileged frame / aether)	4	25	55	19
T3 (local time as artifice)	0	4	1	0

T1 discussion propositions in Einstein 1905 (all 3/3 votes):

plays a part is a conception of synchronism."

common time between A and B by the light-symmetry procedure.

definition is consistent for any number of clocks.

simultaneity has an absolute meaning across frames.

- time_conceptions_are_synchronism – "Every conception in which time
- common_time_by_light_symmetry_definition – the definition of
- synchronism_definition_consistent – the assumption that the
- synchronism_has_no_absolute_significance – the denial that

T2 discussion propositions in Einstein 1905 (all 3/3 votes):

- relative_motion_only
- no_absolute_rest
- no_absolute_space_no_point_velocity
- principle_of_relativity

Use counts: 11 / 12 / 14 propositions tagged as predictions per verifier. Zero of them required T1, T2, or T3 as background. Every prediction required only OTHER (electromagnetic principles, Maxwell transformations, kinematic definitions).

Ratios (use / (discussion+1)):

class	Einstein 1905	1904	1897	1880
T1	0.00	1.25	1.50	4.50
T2	0.00	1.92	0.46	1.00
T3	0.00	0.40	0.50	0.00

Einstein's ratio for every class is zero because his paper contains zero propositions that "use" the class as background for a prediction. Under the ratio measure, Einstein 1905 is not a corpus about T1/T2/T3 at all; it is a corpus that reconstructs the predictions to avoid depending on them.

3. Gate decisions

gate	decision	reason
Q1 extraction sanity	GO	31 propositions $\geq$ 15
Q2 T1 dominates discussion	NO_GO	T1 = 4, T2 = 4 (tie, not strict inequality)
Q3 T1 discussion spike	NO_GO	Einstein T1 disc = 4, 1904 pre-cut T1 disc = 7
Q4 ratio inversion	GO	Einstein T1 ratio = 0.00 < 1.25

Overall NO_GO. The trajectory reframe from DCR3d predicted the discussion spike would appear in Einstein 1905. Under the preregistered raw-count metric it did not.

Licensed reading (from the runner):

t1_not_dominant_in_einstein_1905: Einstein's paper does not discuss T1 more than T2 under our consensus tagging. Would falsify the trajectory framing's specific prediction that Einstein's move IS the T1 discussion spike.

4. What actually happened, in more detail than the gates

Two things surprised us and are worth naming, both in ways the gates do not capture.

4.1 Einstein equalises T1 and T2 discussion

The 1904 pre-cut corpus discusses T2 (aether / privileged frame) 25 times and T1 (simultaneity / common time) 7 times, in aggregate. This section originally read that aggregate as "the precursor era was heavily focused on the aether question, with simultaneity treated as a subtopic." The §4.4 per-doc reslice sharpens this: the aggregate T1 count of 7 is not distributed across the corpus – it is 4 in Poincaré 1898 alone and 1 each in three other papers. The bulk of the corpus does not discuss T1 at all. And the T2 total of 25 is concentrated in several papers (FitzGerald, Larmor, Lodge, Michelson, Rayleigh), none of which also discusses T1.

Einstein's 1905 paper is the first document in the corpus with both T1 and T2 discussion counts at ≥ 4 . He treats T1 and T2 as *the same problem*: two postulates of matched structural form, each denying an absolute quantity.

rest frame – negates T2.

across separated observers except by convention – negates T1.

- The **principle of relativity** denies that there is a privileged
- The **definition of synchronism** denies that there is a common time

The paper's argumentative move is exactly symmetric across T1 and T2. That is a genuinely new datum about what a revolutionary paper looks like: not "the paper that argues most about the deleted commitment," but "the paper that discovers the deletable commitment can be paired with another and both dispatched together."

DCR3d's trajectory framing predicted asymmetric spike on T1 because that was Einstein's specific move. What Einstein actually did was *symmetrise* the two commitments; the per-paper T1-only-or-T2-only asymmetry across the precursor literature (Poincaré 1898 on T1; FitzGerald / Larmor 1900 ch10 / Lodge on T2) is what he removed, not what he acted on.

4.2 Zero prediction-dependence on T1, T2, or T3

Every one of Einstein's ~12 prediction-marked propositions was tagged as requiring only OTHER – electromagnetic principles, Maxwell transformations, kinematic definitions, but not T1, T2, or T3.

This is what the deletion actually looks like *in the text*: the paper's derivations are reconstructed so that no prediction has T1, T2, or T3 as a necessary premise. That is a *stronger* condition than "discuss T1 more." A revolutionary paper does not merely change the mix of what is talked about; it reshapes the derivations so the targeted commitment is no longer a load-bearing beam in any prediction.

Under this measure, Einstein's paper is 100% discussion and 0% use for T1/T2/T3. Every pre-cut paper in the corpus has some prediction- dependence on either T1 or T2. Only Einstein 1905 has zero.

4.3 A per-document normalisation runs the other way

The 1904 pre-cut corpus contains 13 documents (Maxwell 1865 through Lorentz 1904). T1 discussion is 7 propositions across 13 documents = 0.54 T1 discussions per document. Einstein

1905 is one document with 4 T1 discussions = 4.0 T1 discussions per document.

By per-document normalisation, Einstein 1905 has ~7× more T1 discussion per document than the average 1904 document. That measure DOES fit the trajectory framing. But it was not preregistered and should not be counted as a gate result.

The two normalisations tell different stories:

- **Raw count:** the 1904 corpus discusses T1 more (7 vs 4).
- **Per-document:** Einstein 1905 discusses T1 more (4.0 vs 0.54).

Which one you use encodes what "discussion spike" means. The preregistered version was raw count, which said the precursor era owns the discussion. Per-document normalisation would say the revolutionary paper is more intensely focused. Both are true statements about different quantities; neither is the natural one to preregister for the follow-on tests.

4.4 Per-document reslice (correction to §4.1 and §4.3)

Immediately after DCR4 landed, the human director requested a per-document reslice of the DCR3d discussion tags at 1904, on the specific concern that the aggregate 1904 T1 count (7) could be dominated by one paper and the "revolutionary paper is quieter than the precursors" framing might be an averaging artifact. It was.

T1 discussion counts per document at 1904 (≥ 2-of-3 verifier consensus):

document	T1 discussion count
poincare_1898_time	4
poincare_1904_stlouis	1
larmor_1900_ch11	1
lorentz_1904	1
all other 11 documents	0
– total across corpus	7

T2 discussion counts per document at 1904:

document	T2 discussion count
fitzgerald_1889	4
larmor_1900_ch10	4
lodge_1897_absence	3
michelson_1881	3
others (2 each or less)	≤ 2
– total across corpus	25

Einstein 1905 for comparison: T1 = 4, T2 = 4.

Three implications for the paper's original claims:

1905 does not have less T1 discussion than the precursor era; it has the same T1 discussion count as Poincaré 1898 alone (both = 4). The aggregate 7 was one paper of Poincaré's plus three single-mention papers. The paper's original abstract and section 4 framed this as if the precursors collectively created a discussion spike Einstein was quieter than; that framing is not supported by the reslice.

- **The "quieter than the precursors" claim is retracted.** Einstein

4:4 T1:T2 balance is unique in the entire corpus. Poincaré 1898 is 4:0 (T1 alone). FitzGerald 1889 and Larmor 1900 ch10 are 0:4 (T2 alone). No single precursor paper discusses both commitments at ≥ 4 counts each. Einstein does. That is a real per-paper structural signature that survives the reslice.

- **"Symmetric equalisation" claim is strengthened.** Einstein 1905's

paper has zero T1/T2/T3 use in predictions (the raw counts in DCR3c are not zero for any of the 13 documents). Einstein 1905 has zero. This structural property does not depend on aggregation and is independently confirmed by the reslice.

- **"Prediction-independence" claim is unaffected.** No precursor

Net effect on the paper's headline: the "quieter than the precursors" finding is retracted. The two structural moves (equalisation and prediction-independence) are sharpened, not weakened. What the paper actually established is that these two per-paper structural moves are unique to Einstein 1905 in the pre-1905 corpus, and that the aggregate count comparison the original abstract leaned on was misleading.

5. What DCR4 does not license

operationalisation on one specific revolution.

does – it just distributes that content differently from what a discussion-count metric captures.

sixth serial null now (DCR3, DCR3b, DCR3c, DCR3d, DCR3e post-hoc, DCR4 fresh). That is worth naming as a structural finding, not just a bad streak.

papers/dynamics_of_conceptual_deletion/paper.md (in flight, being drafted concurrently by a separate agent). DCR4's finding strengthens some claims there – the discussion spike being in the precursor era, not the paper itself, is directly what that paper reframes toward – and pushes back on others (equalisation and prediction-independence are new structural moves that DCD's framework does not yet name).

- The trajectory framing is refuted in general. It failed one specific
- Einstein 1905 has no argumentative content about T1/T2. It clearly
- The DR-arc program is complete. The methodology has produced a
- The DCD companion paper's framework is validated. See

6. What comes next

The DR-arc as a scoring-based nominator is done. Six preregistered nulls in a row with the target commitment recoverable *after* the fact from post-hoc reweighting and never *before* the fact from a committed procedure. The structural claim that survives is DR5's verification-limit theorem: proposition-ranking nominators cannot distinguish D from any of its realisations when the deletion decomposes into multiple concrete propositions.

Directions that would extend this correctly:

tectonics, quantum mechanics. The DCD companion paper proposes this. What DCR4 adds: test not just for a T1-analogue discussion spike but for the two structural signatures found here – *equalisation* of discussion across previously asymmetric commitments, and *prediction-independence* of the paper's derivations from the deleted commitment.

literature consistently discusses the eventually-deleted commitment *more* than the revolutionary paper does. If so, the discussion spike is a precursor phenomenon, not a revolutionary-paper phenomenon – and the "detect an incoming revolution" application should look at literature dynamics, not any single paper.

- **Multi-case extension** on Copernicus, Darwin, Lavoisier, plate
- **Precursor-vs-revolution split.** Test whether the precursor

normalisations in §4.3 tell opposite stories on the same data. Any further work must decide which normalisation is the natural one for the question being asked. For "does the community discuss X more?" raw count is right. For "is this paper unusually focused on X?" per-document

(or per-word) is right. The DR-arc conflated them.

- **Retire raw counts, use per-corpus-mass normalisation.** The two

Appendix: reproduction

```
# Fetch and cache Einstein 1905
```

```
uv run --no-sync python -c "from experiments.date_cut_retrodiction.fetch import _fetch_plain, DATA_DIR; _, t = _fetch_plain"
```

```
# Consensus + score (given tagger outputs already present)
```

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr4
```

Extraction and tagging use nine sandboxed subagent invocations (3 extraction + 3 use + 3 discussion) with the DCR1e / DCR3c / DCR3d prompts. Each subagent reads exactly one file and writes one JSON.

Preregistration **digest** **(SHA-256** **of** DCR4_PREREGISTRATION.md):
5907269c61681008273c66ee71b96faa456e5191ad2c7258723372550ef7211c.

Extraction **prompt** **digest** **(SHA-256** **of** EXTRACTION_PROMPT PRESUPPOSITION.md):
a3e01ef0d6793cdc16692f7e5ec03e75ef67c326e277460490e33f21eed64b61.

Discussion **prompt** **digest** **(SHA-256** **of** DCR3D_PROMPT.md):
24384377bfab8dfe87cc72ce01d0908da992cddb2e39d11781cb42f432dbea05.